

Use Spotify Audio Features To Categorize Music

Practical Exam

Goal

Spotify provides an API for basic track information:

- **`ID`**, e.g. *2lEcSduKEXEK5KJ9hJzlCz*
- **`name`**, e.g. *Gloana Bauer (Teenage Dirtbag)*
- **`artist`**, e.g. *D‘ Hundskrippln*
- *…*

as well as related audio features:

- **`energy`**, e.g. *0.454*
  Energy is a measure from 0.0 to 1.0 and represents a perceptual measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy. For example, death metal has high energy, while a Bach prelude scores low on the scale.
- **`speechiness`**, e.g. *0.0388*
  Speechiness detects the presence of spoken words in a track. The more exclu sively speech-like the recording (e.g. talk show, audio book, poetry), the closer to 1.0 the attribute value.
- **`loudness`**, e.g. *-8.758*
  The overall loudness of a track in decibels (dB). Loudness values are averaged.
- **`tempo`**, e.g. *97.532*
  The overall estimated tempo of a track in beats per minute (BPM)
- **`acousticness`**, *e.g. 0.268*
  A confidence measure from 0.0 to 1.0 of whether the track is acoustic.
- *…*

https://developer.spotify.com/documentation/web-api/reference/#/operations/get-audio-features



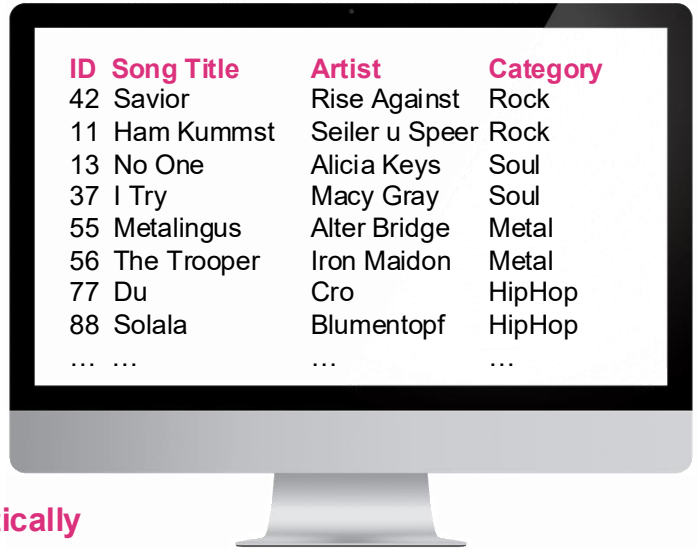
Goal

We want to make use of those audio features to automatically assign each track to a certain category:

Metal	***Classic***	***Rock***	***Vocal***
Electro	***Podcast***	***Soul***	***HipHop***

Workflow:

- **Query data** from Spotify API
- **Save raw data** (*JSON* files) to HDFS
- **Optimize**, **reduce** and **clean raw data** and save it to **final** directory on HDFS
- **Calculate categories** (*Metal, Classic, Rock, …*)
- **Join track information** and **audio festures** and **save** everything **to end-user database** (e.g. MySQL, MongoDB…)
- Provide a simple HTML Frontend which reads from end-user database and displays result
- The whole data workflow **must be implemented** within an ETL **workflow tool** (e.g. Pentaho Data Integration or Airflow) and **run automatically**



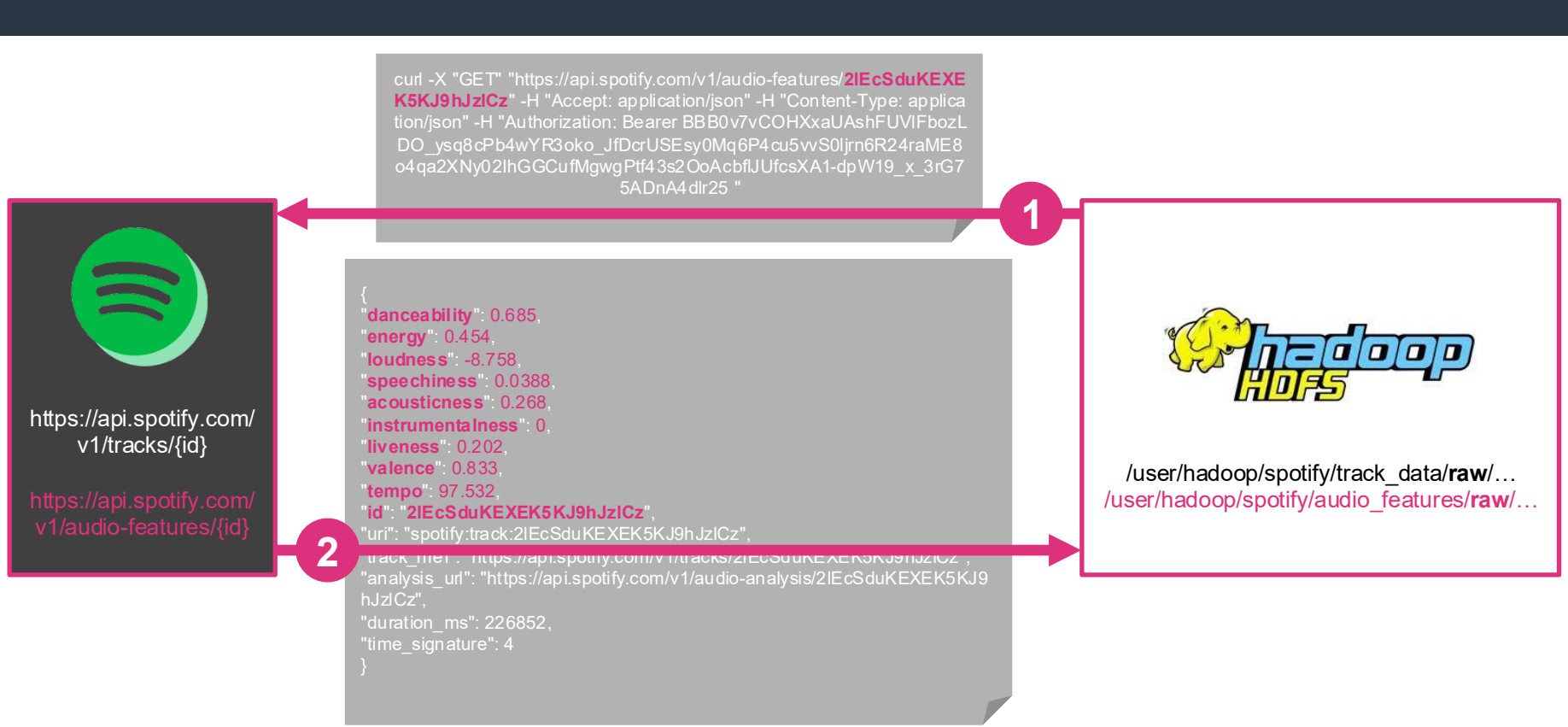
ID	Song Title	Artist	Category
42	Savior	Rise Against	Rock
11	Ham Kummst	Seiler u Speer	Rock
13	No One	Alicia Keys	Soul
37	I Try	Macy Gray	Soul
55	Metalingus	Alter Bridge	Metal
56	The Trooper	Iron Maidon	Metal
77	Du	Cro	HipHop
88	Solala	Blumentopf	HipHop
…	…	…	…

Dataflow: 1. Get Track Information



https://developer.spotify.com/documentation/web-api/reference/#/operations/get-track

Dataflow: 2. Get Track Audio Features



https://developer.spotify.com/documentation/web-api/reference/#/operations/get-audio-features

Dataflow: 1. Get Track Information

```
curl -X "GET" " https://api.reccobeats.com/v1/track/search?searchText
=kavinsky"
```



https://api.reccobeats.com/v1/track/search?searchText=kavinsky

https://api.reccobeats.com/v1/track/{id}/audio-features

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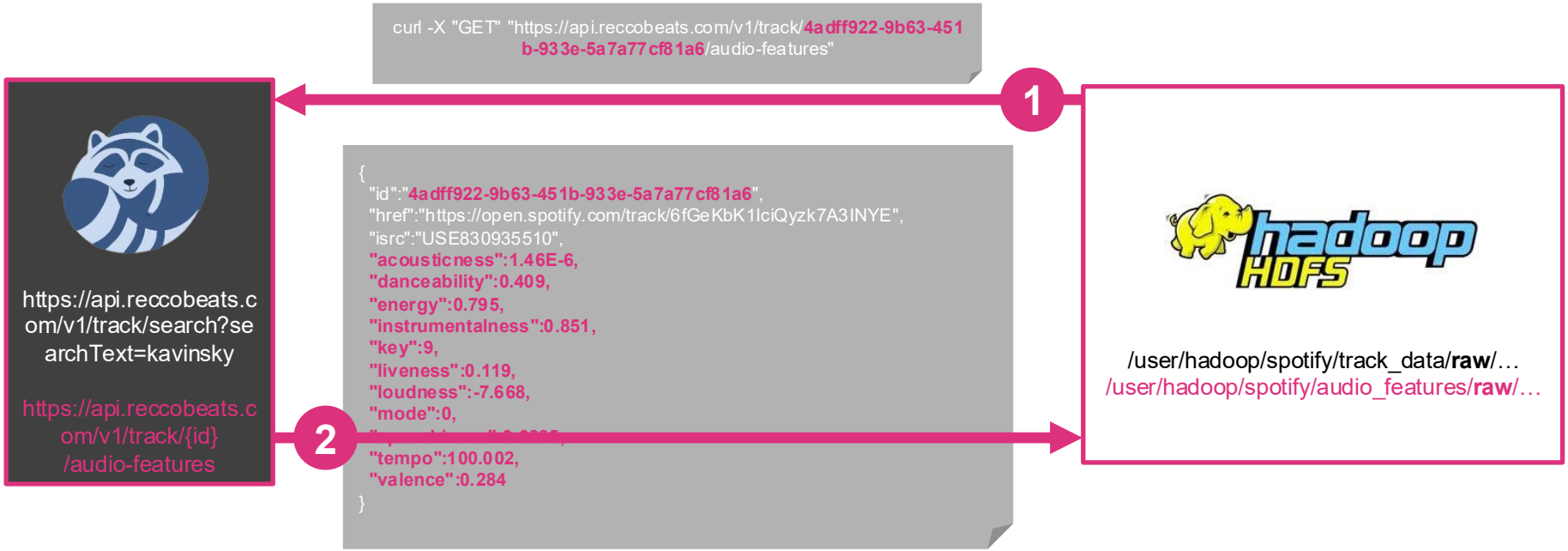

```
[…]
{
  "id":"4adff922-9b63-451b-933e-5a7a77cf81a6",
  "trackTitle":"Nightcall (Kavinsky)",
  "artists":[{
      "id":"b5bfcf4a-8a95-4ed6-955e-6a1b44709da0",
      "name":"Charlie Parra del Ri ego",
      "href":"https://open.spotify.com/artist/31Eoi17HcQM7ElqMMXIUU3"
  }],
  "durationMs":215624,
  "isrc":"USE830935510",
  "ean":null,
  "upc":null,
  "href":"https://open.spotify.com/track/6fGeKbK1IciQyzk7A3INYE",
  "availableCountries":"AR,AU,AT,BE,BO,BR,BG,CA,CL,CO,CR,CY,CZ,DK,DO,DE,EC,EE,SV,FI,FR,GR,GT,HN,HK,HU,IS,IE,IT,LV,LT,LU,MY,MT,MX,NL,NZ,NI,NO,PA,PY,PE,PH,PL,PT,SG,SK,ES,SE,CH,TW,TR,UY,US,GB,AD,LI,MC,ID,JP,TH,VN,RO,IL,ZA,SA,AE,BH,QA,OM,KW,EG,MA,DZ,TN,LB,JO,PS,IN,BY,KZ,MD,UA,AL,BA,HR,ME,MK,RS,SI,KR,BD,PK,LK,GH,KE,NG,TZ,UG,AG,AM,BS,BB,BZ,BT,BW,BF,CV,CW,DM,FJ,GM,GE,GD,GW,GY,HT,JM,KI,LS,LR,MW,MV,ML,MH,FM,NA,NR,NE,PW,PG,PR,WS,SM,ST,SN,SC,SL,SB,KN,LC,VC,SR,TL,TO,TT,TV,VU,AZ,BN,BI,KH,CM,TD,KM,GQ,SZ,GA,GN,KG,LA,MO,MR,MN,NP,RW,TG,UZ,ZW,BJ,MG,MU,MZ,AO,CI,DJ,ZM,CD,CG,IQ,LY,TJ,VE,ET,XK",
  "popularity":5
}
[…]
```




/user/hadoop/spotify/track_data/**raw**/…
/user/hadoop/spotify/audio_features/**raw**/…

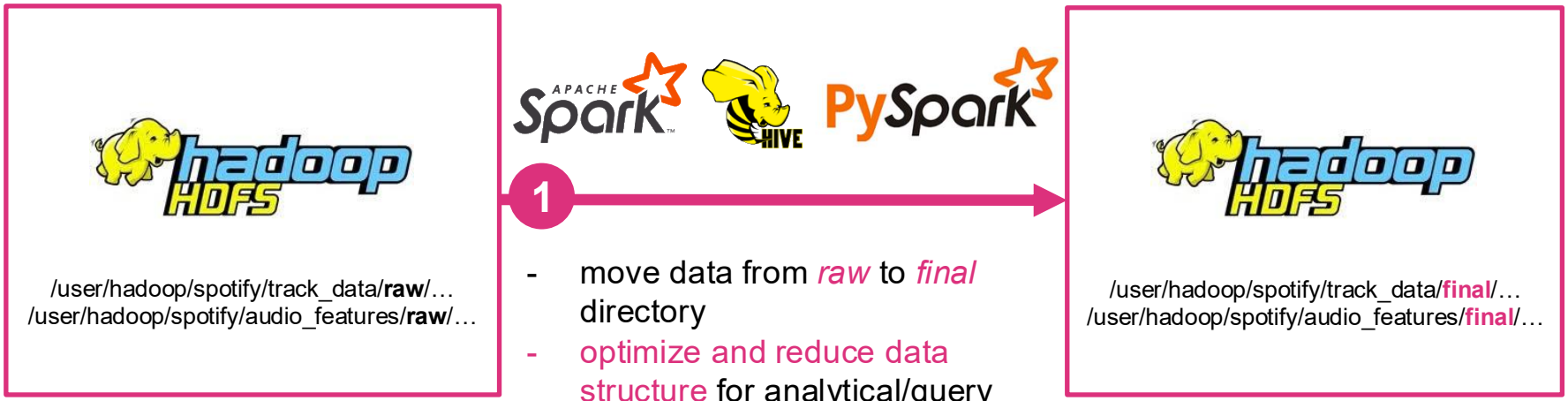
https://reccobeats.com/docs/apis/search-tracks

Dataflow: 2. Get Track Audio Features

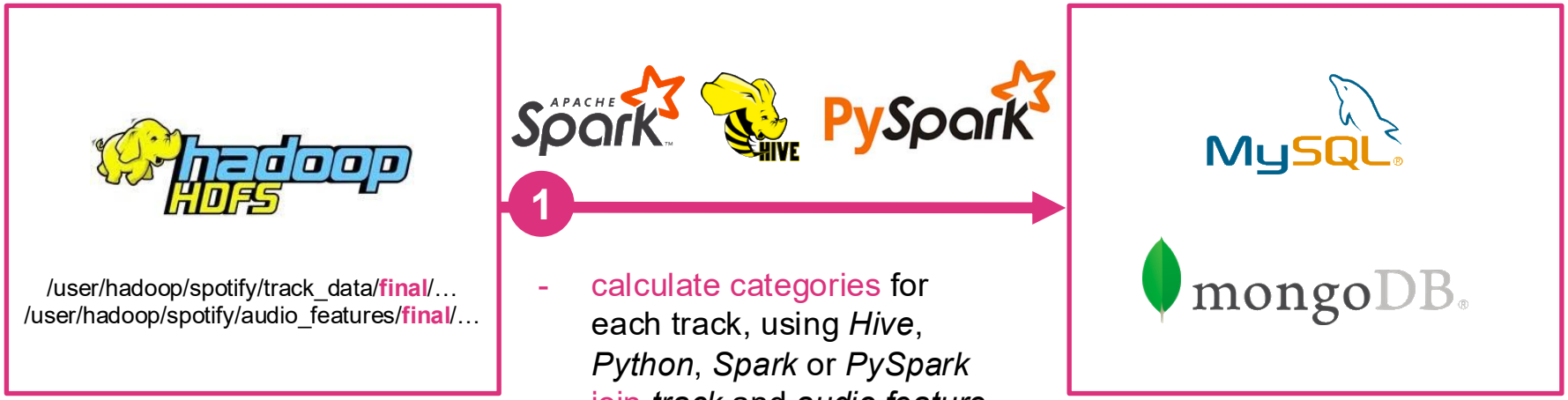


https://developer.spotify.com/documentation/web-api/reference/#/operations/get-audio-features

Dataflow: 3. Raw To Final Transfer



Dataflow: 4. Run Analysis and Save Results



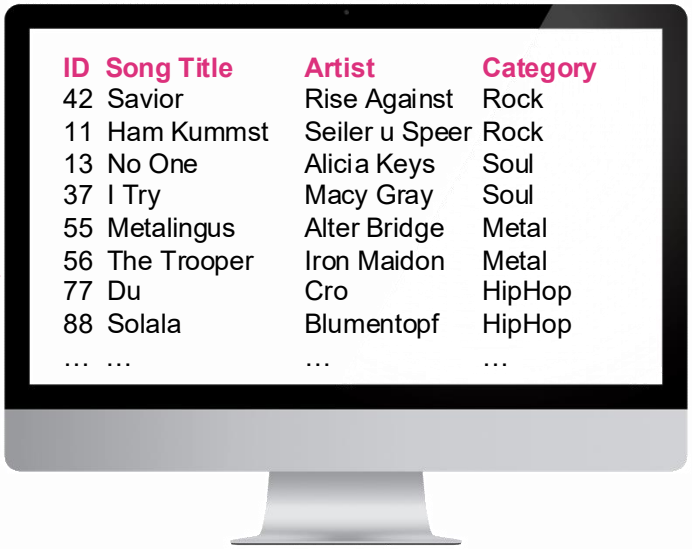
- calculate categories for each track, using *Hive*, *Python*, *Spark* or *PySpark*
- join *track* and *audio feature* data
- save everything to a end-user database (e.g. *MySQL*, *MongoDB*)

Dataflow: **5. Provide Simple Web Interface**



1

- Create a simple Website which reads and displays data from end-user database



ID	Song Title	Artist	Category
42	Savior	Rise Against	Rock
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